

Overview of Protein structure prediction

Subject: Protein structure prediction

1.

Other improvements It is reported that in addition to the protein sequence, secondary structure formation depends on other factors. For example, it is reported that secondary structure tendencies depend also on local environment, solvent accessibility of residues, protein structural class, and even the organism from which the proteins are obtained. Based on such observations, some studies have shown that secondary structure prediction can be improved by addition of information about protein structural class, residue accessible surface area and also contact number information.

2.

Current AI methods and databases of predicted protein structures AlphaFold2, was introduced in CASP14, and is capable of predicting protein structures to near experimental accuracy. AlphaFold was swiftly followed by RoseTTAFold and later by OmegaFold and the ESM Metagenomic Atlas. In a study, Sommer et al. 2022 demonstrated the application of protein structure prediction in genome annotation, specifically in identifying functional protein isoforms using computationally predicted structures, available at <https://www.isoform.io>. This study highlights the promise of protein structure prediction as a genome annotation tool and presents a practical, structure-guided approach that can be used to enhance the annotation of any genome. In 2024, David Baker and Demis Hassabis (along with John M. Jumper) were awarded the Nobel Prize in Chemistry for their contributions to computational protein modeling, including the development of AlphaFold2, an AI-based model for protein structure prediction. AlphaFold2's accuracy has been evaluated against experimentally determined protein structures using metrics such as root-mean-square deviation (RMSD). The median RMSD between different experimental structures of the same protein is approximately 0.6 Å, while the median RMSD between AlphaFold2 predictions and experimental structures is around 1 Å. For regions where AlphaFold2 assigns high confidence, the median RMSD is about 0.6 Å, comparable to the variability observed between different experimental structures. However, in low-confidence regions, the RMSD can exceed 2 Å, indicating greater deviations. In proteins with multiple domains connected by flexible linkers, AlphaFold2 predicts individual domain structures accurately but may assign random relative positions to these domains. Additionally, AlphaFold2 does not account for structural constraints such as the membrane plane, sometimes placing protein domains in positions that would physically clash with the membrane.

3.

In the case of complexes of two or more proteins, where the structures of the proteins are known or can be predicted with high accuracy, protein–protein docking methods can be used to predict the structure of the complex. Information of the effect of mutations at specific sites on the affinity of the complex helps to understand the complex structure and to guide docking methods.

4.

Terms used for classifying protein structures and sequences The more commonly used terms for evolutionary and structural relationships among proteins are listed below. Many additional terms are

used for various kinds of structural features found in proteins. Descriptions of such terms may be found at the CATH Web site, the Structural Classification of Proteins (SCOP) Web site, and a Glaxo Wellcome tutorial on the Swiss bioinformatics Expasy Web site.

5.

Protein structure prediction is the inference of the three-dimensional structure of a protein from its amino acid sequence—that is, the prediction of its secondary and tertiary structure from primary structure. Structure prediction is different from the inverse problem of protein design. Protein structure prediction is one of the most important goals pursued by computational biology and addresses Levinthal's paradox. Accurate structure prediction has important applications in medicine (for example, in drug design) and biotechnology (for example, in novel enzyme design). Starting in 1994, the performance of current methods is assessed biennially in the Critical Assessment of Structure Prediction (CASP) experiment. A continuous evaluation of protein structure prediction web servers is performed by the community project Continuous Automated Model EvaluatiOn (CAMEO3D).

6.

Before modelling Most tertiary structure modelling methods, such as Rosetta, are optimized for modelling the tertiary structure of single protein domains. A step called domain parsing, or domain boundary prediction, is usually done first to split a protein into potential structural domains. As with the rest of tertiary structure prediction, this can be done comparatively from known structures or ab initio with the sequence only (usually by machine learning, assisted by covariation). The structures for individual domains are docked together in a process called domain assembly to form the final tertiary structure.

7.

Comparative protein modeling Comparative protein modeling uses previously solved structures as starting points, or templates. This is effective because it appears that although the number of actual proteins is vast, there is a limited set of tertiary structural motifs to which most proteins belong. It has been suggested that there are only around 2,000 distinct protein folds in nature, though there are many millions of different proteins. The comparative protein modeling can combine with the evolutionary covariation in the structure prediction. These methods may also be split into two groups:

8.

Modeling of side-chain conformations Accurate packing of the amino acid side chains represents a separate problem in protein structure prediction. Methods that specifically address the problem of predicting side-chain geometry include dead-end elimination and the self-consistent mean field methods. The side chain conformations with low energy are usually determined on the rigid polypeptide backbone and using a set of discrete side chain conformations known as "rotamers". The methods attempt to identify the set of rotamers that minimize the model's overall energy. These methods use rotamer libraries, which are collections of favorable conformations for each residue type in proteins. Rotamer libraries may contain information about the conformation, its frequency, and the standard deviations about mean dihedral angles, which can be used in sampling. Rotamer libraries are derived from structural bioinformatics or other statistical analysis of side-chain conformations in known experimental structures of proteins, such as by clustering the observed conformations for tetrahedral carbons near the staggered (60° , 180° , -60°) values. Rotamer libraries can be

backbone-independent, secondary-structure-dependent, or backbone-dependent. Backbone-independent rotamer libraries make no reference to backbone conformation, and are calculated from all available side chains of a certain type (for instance, the first example of a rotamer library, done by Ponder and Richards at Yale in 1987). Secondary-structure-dependent libraries present different dihedral angles and/or rotamer frequencies for α $\{\displaystyle \alpha\}$ -helix, β $\{\displaystyle \beta\}$ -sheet, or coil secondary structures. Backbone-dependent rotamer libraries present conformations and/or frequencies dependent on the local backbone conformation as defined by the backbone dihedral angles ϕ $\{\displaystyle \phi\}$ and ψ $\{\displaystyle \psi\}$, regardless of secondary structure. The modern versions of these libraries as used in most software are presented as multidimensional distributions of probability or frequency, where the peaks correspond to the dihedral-angle conformations considered as individual rotamers in the lists. Some versions are based on very carefully curated data and are used primarily for structure validation, while others emphasize relative frequencies in much larger data sets and are the form used primarily for structure prediction, such as the Dunbrack rotamer libraries. Side-chain packing methods are most useful for analyzing the protein's hydrophobic core, where side chains are more closely packed; they have more difficulty addressing the looser constraints and higher flexibility of surface residues, which often occupy multiple rotamer conformations rather than just one.